

Two-coin counterexamples to the proposed generalized Shepp–Olkin thresholds

All finite orders above one, exact rational certificates,
and the fixed-mean entropy phase diagram

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Abstract

Hillion and Johnson proposed that the Rényi entropy of a sum of independent Bernoulli variables is jointly concave in the Bernoulli parameters up to order 2, and that the corresponding Tsallis threshold is approximately 3.65986. We give a counterexample to both proposed numerical thresholds. In fact, for every finite real $q > 1$, both entropies fail even quasiconcavity for the sum of *two* independent Bernoulli variables, with all parameters strictly between zero and one. The obstruction is a mean-preserving interpolation $(p + t, p - t)$ near the all-zero corner. Its power sum has a negative second variation at the balanced point when p is sufficiently small.

The counterexamples admit a fully explicit dyadic family. At $q = 2$, one particularly small certificate has midpoint $(1/10, 1/10)$ and endpoints $(3/20, 1/20)$ and $(1/20, 3/20)$; the Tsallis midpoint deficit is exactly $181/80000$. We also determine the sharp balanced-point curvature transition, classify every fixed-mean two-coin entropy maximizer for $q > 1$, prove concavity on all two-coin fixed-mean segments for $0 < q \leq 1$, and identify the minimum number of coins needed for failure. The transition near $q = 1$ occurs on an exponentially small parameter scale.

The all-order disproof is elementary and does not depend on numerical experiments. Reproducible exact and symbolic checks accompany the article. The general joint-concavity question for $0 < q < 1$ is not settled here, and bibliographic priority and the proofs have not been independently refereed.

Selection record. A single OS-backed uniform draw selected *Discrete probability*, area 58 of 96. The supplied manifest’s 71 excluded research packages were reviewed before the problem was chosen. The complete area list and original draw are included in the archive.

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1 The published problem and the scope of the result

1.1 Bernoulli sums and entropy

Let $X_1, \dots, X_n$ be independent, with $\mathbb{P}(X_i = 1) = p_i$ and $\mathbb{P}(X_i = 0) = 1 - p_i$. Write

$$S_{\mathbf{p}} = X_1 + \dots + X_n, \quad f_k(\mathbf{p}) = \mathbb{P}(S_{\mathbf{p}} = k), \quad \mathbf{p} \in [0, 1]^n. \quad (1.1)$$

The distribution is often called Poisson–binomial. Throughout this article, q is a *finite real order*, logarithms are natural, and $0^q = 0$ for $q > 0$. Define the power sum and the two entropies by

$$F_q(\mathbf{p}) = \sum_{k=0}^n f_k(\mathbf{p})^q, \quad (1.2)$$

$$H_{R,q}(\mathbf{p}) = \frac{\log F_q(\mathbf{p})}{1 - q}, \quad H_{T,q}(\mathbf{p}) = \frac{1 - F_q(\mathbf{p})}{q - 1}, \quad q \neq 1. \quad (1.3)$$

Both approach the Shannon entropy $H_1(\mathbf{p}) = -\sum_k f_k(\mathbf{p}) \log f_k(\mathbf{p})$ as $q \rightarrow 1$. For $q > 1$, both functions in (1.3) are strictly *decreasing* functions of F_q .

Joint concavity means

$$H((1 - \theta)\mathbf{x} + \theta\mathbf{y}) \geq (1 - \theta)H(\mathbf{x}) + \theta H(\mathbf{y}) \quad (0 \leq \theta \leq 1). \quad (1.4)$$

It tests every affine path in the parameter cube, including paths on which some coordinates increase and others decrease. It is not the same as concavity along the diagonal $p_1 = \dots = p_n$, and it is not the same as concavity in each coordinate separately.

1.2 The target

Hillion and Johnson proved joint concavity for Shannon entropy [1, Theorem 1.2]. The target here is the *proposed numerical threshold assertion adjoining Conjecture 4.2* of that paper’s arXiv version: the proposed upper orders are 2 for Rényi entropy and 3.65986... for Tsallis entropy. The latter is the root greater than one of $2 - 4q + 2^q = 0$. Johnson repeated these proposals in his survey’s open-problems section [2, Section 8, item 5]. The statement is about all Bernoulli parameter interpolations, not just monotone ones.

Theorem 1.1 (Main disproof). *For every finite real $q > 1$, each of the maps*

$$(p_1, p_2) \mapsto H_{R,q}(p_1, p_2), \quad (p_1, p_2) \mapsto H_{T,q}(p_1, p_2)$$

fails quasiconcavity on $(0, 1)^2$, and therefore fails joint concavity there. Counterexamples can be chosen on a fixed-mean affine segment contained in $(0, 1/2)^2$, with rational, indeed dyadic, coordinates.

The distinction between *threshold values* and *existence of a threshold interval* matters. Theorem 1.1 refutes the proposed values, but does not by itself establish concavity for every $0 < q < 1$. If the initial-interval formulation of Conjecture 4.2 is correct, its finite threshold must instead be 1. Equivalently, because the Shannon case is known, the largest finite order for which universal joint concavity can hold is exactly 1; the full set of admissible orders below 1 is not determined here.

1.3 Source and novelty audit

The original arXiv PDF, including page 7 with the conjecture, was inspected. The survey independently confirms the proposed constants. The later monotonicity paper concerns a different property [3], while the Bernoulli-sum Rényi inequalities of Madiman, Melbourne and Roberto

concern variance, convolution and entropy-power comparisons [4]. No earlier resolution of the specific numerical-threshold claim was located in the searches performed on 20 September 2026. This is a bounded search result, not a guarantee of priority. The exact theorem proved below is independent of whether the same counterexample has appeared elsewhere. A detailed source and search audit is included in `notes/sources.md`.

2 A complete rational counterexample at order two

Before proving an all-order theorem, it is useful to give a certificate that requires only arithmetic with three probabilities.

Take the two endpoint parameter vectors and their midpoint

$$\mathbf{x} = \left(\frac{3}{20}, \frac{1}{20}\right), \quad \mathbf{y} = \left(\frac{1}{20}, \frac{3}{20}\right), \quad \mathbf{m} = \frac{\mathbf{x} + \mathbf{y}}{2} = \left(\frac{1}{10}, \frac{1}{10}\right). \quad (2.1)$$

The endpoint distributions agree because exchanging the two independent coins does not change their sum. Direct multiplication gives

$$f(\mathbf{x}) = f(\mathbf{y}) = \frac{1}{400}(323, 74, 3), \quad f(\mathbf{m}) = \frac{1}{100}(81, 18, 1). \quad (2.2)$$

Every probability and every Bernoulli parameter is positive. Both sums have mean $1/5$. Their collision probabilities are

$$F_2(\mathbf{m}) = \frac{81^2 + 18^2 + 1}{10000} = \frac{55088}{80000}, \quad (2.3)$$

$$F_2(\mathbf{x}) = F_2(\mathbf{y}) = \frac{323^2 + 74^2 + 3^2}{400^2} = \frac{54907}{80000}. \quad (2.4)$$

Thus the midpoint has *larger* collision probability, by $181/80000$. Since $H_{T,2} = 1 - F_2$, its Jensen deficit is exactly

$$\frac{H_{T,2}(\mathbf{x}) + H_{T,2}(\mathbf{y})}{2} - H_{T,2}(\mathbf{m}) = \frac{181}{80000} > 0. \quad (2.5)$$

For Rényi entropy, $H_{R,2} = -\log F_2$, so the corresponding deficit is

$$\log \frac{55088}{54907} > 0, \quad \text{approximately } 0.003291061655. \quad (2.6)$$

The approximate decimal is illustrative; strict positivity follows from the integer inequality $55088 > 54907$.

This one example already refutes both proposed thresholds: the order 2 is in each proposed concavity range. It also shows why concavity in the probability vector is irrelevant. Although $\mathbf{m}$ is the midpoint of the *coin parameters*, $f(\mathbf{m})$ is not the midpoint of the endpoint probability vectors; that midpoint is simply $f(\mathbf{x})$.

3 Counterexamples for every finite order above one

3.1 The mean-preserving two-coin path

Fix $0 < p \leq 1/2$ and consider

$$p_1(t) = p + t, \quad p_2(t) = p - t, \quad |t| \leq p. \quad (3.1)$$

The three probabilities are

$$f_0(t) = (1 - p)^2 - t^2, \quad f_1(t) = 2p(1 - p) + 2t^2, \quad f_2(t) = p^2 - t^2. \quad (3.2)$$

Their sum is one and their mean is $2p$ for every t . Let

$$A = (1 - p)^2, \quad B = 2p(1 - p), \quad C = p^2, \quad a = q - 1. \quad (3.3)$$

The path is even in t . Writing $u = t^2$ isolates its essential dependence:

$$G_{q,p}(u) = (A - u)^q + (B + 2u)^q + (C - u)^q, \quad 0 \leq u \leq p^2, \quad (3.4)$$

so that $F_q(p + t, p - t) = G_{q,p}(t^2)$. For $q > 1$,

$$G'_{q,p}(u) = q[-(A - u)^a + 2(B + 2u)^a - (C - u)^a]. \quad (3.5)$$

Lemma 3.1 (A sufficient small-mean condition). *Suppose $q > 1$ and*

$$0 < p < \frac{1}{2(1 + 2^{1/(q-1)})}. \quad (3.6)$$

Then $G_{q,p}$ is strictly decreasing on $[0, p^2]$. Consequently, for every $0 < |t| \leq p$, both entropies in (1.3) are strictly larger at $(p + t, p - t)$ than at (p, p) .

Proof. On the stated interval,

$$A - u \geq 1 - 2p, \quad B + 2u \leq 2p, \quad C - u \geq 0.$$

Because $a > 0$, (3.5) gives

$$G'_{q,p}(u) \leq q[-(1 - 2p)^a + 2(2p)^a] < 0.$$

The strict inequality is exactly (3.6). The power function is continuously differentiable at zero for $q > 1$, so this argument also applies at the endpoint. The entropy conclusion follows because both entropies strictly decrease as functions of F_q . $\square$

3.2 An explicit dyadic construction

Theorem 3.2 (Uniform constructive witness). *For $q > 1$, set*

$$m = 3 + \left\lceil \frac{1}{q - 1} \right\rceil, \quad p = 2^{-m}. \quad (3.7)$$

The points

$$\mathbf{x}_q = (3p/2, p/2), \quad \mathbf{y}_q = (p/2, 3p/2), \quad \mathbf{m}_q = (p, p) \quad (3.8)$$

satisfy, for $H = H_{R,q}$ and for $H = H_{T,q}$,

$$H(\mathbf{m}_q) < H(\mathbf{x}_q) = H(\mathbf{y}_q). \quad (3.9)$$

Proof. Write $a = q - 1 > 0$. Since $m \geq 4$, we have $p \leq 1/16 < 1/4$ and therefore

$$2 \left(\frac{2p}{1 - 2p} \right)^a \leq 2(4p)^a = 2^{1+a(2-m)} \quad (3.10)$$

$$= 2^{1-a(1+\lceil 1/a \rceil)} \leq 2^{-a} < 1. \quad (3.11)$$

Hence $(1 - 2p)^a > 2(2p)^a$, the condition of Lemma 3.1. Apply that lemma with $t = p/2$. The equality between the two endpoints follows from symmetry. All their coordinates lie strictly between zero and $1/2$. $\square$

A reparametrization makes the source's unit-time convention explicit:

$$\mathbf{p}(\theta) = (p(1/2 + \theta), p(3/2 - \theta)), \quad 0 \leq \theta \leq 1.$$

This is an affine Bernoulli interpolation whose slopes are p and $-p$. The midpoint $\theta = 1/2$ violates (1.4).

Proof of Theorem 1.1. Quasiconcavity would require $H((\mathbf{x} + \mathbf{y})/2) \geq \min\{H(\mathbf{x}), H(\mathbf{y})\}$. Equation (3.9) contradicts this weaker inequality. $\square$

Corollary 3.3 (Monotone reparametrizations cannot repair the witness). *Any strictly increasing scalar transform of either entropy still fails quasiconcavity on the same segment. More generally, for fixed $q > 1$, every strictly decreasing function of F_q has the same strict midpoint defect.*

Proof. A strictly monotone transformation preserves the strict ordering and the equality of the two endpoint values in (3.9). $\square$

3.3 Why the construction is genuinely all-order

The proof does not sample q , rely on a graph, or assume that q is an integer. Its only order-dependent ingredient is the monotonicity of x^{q-1} for $q > 1$. For rational $a = A_0/B_0 > 0$ in lowest terms, the sufficient condition for the dyadic witness can also be verified by the integer test

$$(2^m - 2)^{A_0} > 2^{A_0+B_0}. \quad (3.12)$$

This follows by raising $(1-2p)^a > 2(2p)^a$ to the positive integer power B_0 and clearing denominators. The archive records such exact certificates at orders as close to one as 1001/1000. These finite checks are a useful independent check on the formula, but the preceding inequality proof covers all real orders.

4 The exact balanced-point curvature transition

The dyadic construction deliberately uses a conservative small value of p . A sharp local analysis both enlarges the counterexample region and explains the behavior near the Shannon order.

Definition 4.1. For finite $q > 0$ and $0 < p \leq 1/2$, let

$$D_q(p) = (1-p)^{2(q-1)} - 2(2p(1-p))^{q-1} + p^{2(q-1)}. \quad (4.1)$$

At $t = 0$, all three first derivatives in (3.2) vanish. Their second derivatives are $-2, 4, -2$. Thus

$$\left. \frac{d^2}{dt^2} F_q(p+t, p-t) \right|_{t=0} = -2qD_q(p), \quad F'_q(0) = 0. \quad (4.2)$$

Applying the chain rule to (1.3) yields

$$\left. \frac{d^2}{dt^2} H_{T,q}(p+t, p-t) \right|_{t=0} = \frac{2q}{q-1} D_q(p), \quad (4.3)$$

$$\left. \frac{d^2}{dt^2} H_{R,q}(p+t, p-t) \right|_{t=0} = \frac{2q}{(q-1)F_q(p,p)} D_q(p). \quad (4.4)$$

The logarithmic chain-rule square term is zero here. The equality of signs in these two formulas is a special feature of this stationary path, not a general equivalence between Rényi and Tsallis concavity.

Theorem 4.2 (Sharp curvature boundary). *For $q > 1$, write $a = q - 1$ and define*

$$p_c(q) = \frac{1}{1 + \exp(\operatorname{arcosh}(2^a)/a)}. \quad (4.5)$$

Along (3.1), the balanced point $t = 0$ is a strict local entropy minimum when $0 < p < p_c(q)$, and a strict local entropy maximum when $p_c(q) < p \leq 1/2$. At $p = p_c(q)$ the second derivative vanishes, but the balanced point remains a strict local maximum with a negative quartic term. These statements hold for both entropies.

Proof. Let $x = p/(1 - p) \in (0, 1]$ and $y = x^a$. Then

$$D_q(p) = (1 - p)^{2a}(y^2 - 2^{a+1}y + 1). \quad (4.6)$$

The roots of the quadratic are

$$y_- = 2^a - \sqrt{2^{2a} - 1}, \quad y_+ = 2^a + \sqrt{2^{2a} - 1},$$

with $0 < y_- < 1 < y_+$ and $y_- y_+ = 1$. Hence, in the range $0 < y \leq 1$, the sign is positive exactly when $y < y_-$. Since $y_- = \exp(-\operatorname{arcosh}(2^a))$, this is equivalent to $p < p_c(q)$. Equations (4.3)–(4.4) prove the nonzero curvature statements.

For the zero-curvature case, all A, B, C are strictly positive. Taylor expansion gives

$$G_{q,p}(t^2) = G_{q,p}(0) - qD_q(p)t^2 + \frac{q(q-1)}{2}E_q(p)t^4 + O(t^6), \quad (4.7)$$

where

$$E_q(p) = A^{q-2} + 4B^{q-2} + C^{q-2} > 0.$$

When $D_q(p) = 0$, the entropy differences are respectively

$$-\frac{q}{2}E_q(p)t^4 + O(t^6), \quad -\frac{qE_q(p)}{2F_q(p,p)}t^4 + O(t^6).$$

Both are negative for sufficiently small nonzero t . □

Example 4.3 (Order two). Here $D_2(p) = 1 - 6p + 6p^2$ and

$$p_c(2) = \frac{3 - \sqrt{3}}{6} = 0.2113248654\dots \quad (4.8)$$

At $p = 1/10$, the Tsallis directional second derivative is $46/25 > 0$. The full Hessian there is

$$\nabla^2 H_{T,2}(1/10, 1/10) = \frac{1}{25} \begin{pmatrix} -73 & -96 \\ -96 & -73 \end{pmatrix}. \quad (4.9)$$

Its symmetric and antisymmetric eigenvalues are $-169/25$ and $23/25$. Thus both coordinate directions and the diagonal direction have negative curvature, while the mean-preserving direction $(1, -1)$ has positive curvature. Testing only the former directions misses the obstruction.

5 All fixed-mean maximizers for two coins

We now classify the maximizers, not just the curvature at the balanced point. The classification applies simultaneously to the two entropies because their maxima correspond to the same minima of F_q .

5.1 Reduction and the two boundaries

Complementing both Bernoulli variables replaces the sum S by $2 - S$ and reverses its probability vector, without changing either entropy. Therefore it suffices to treat total mean $s \in [0, 1]$. Put $p = s/2$. Every pair of parameters with this mean has the form $(p + t, p - t)$ for $|t| \leq p$.

Define, in addition to (4.5),

$$p_b(q) = \frac{1}{2(1 + 2^{1/(q-1)})}. \quad (5.1)$$

The subscript b refers to the boundary of the parameter segment, and the subscript c to its center.

Lemma 5.1 (Strict convexity in squared imbalance). *For $q > 1$ and $0 < p \leq 1/2$, $G_{q,p}$ is strictly convex on $[0, p^2]$. Its endpoint derivatives are*

$$G'_{q,p}(0) = -qD_q(p), \quad (5.2)$$

$$G'_{q,p}(p^2) = q[2(2p)^{q-1} - (1 - 2p)^{q-1}]. \quad (5.3)$$

Moreover $0 < p_b(q) < p_c(q) < 1/2$.

Proof. In the interior,

$$G''_{q,p}(u) = q(q-1)[(A-u)^{q-2} + 4(B+2u)^{q-2} + (C-u)^{q-2}] > 0. \quad (5.4)$$

Continuity extends strict convexity to the closed interval. The endpoint formulas follow from (3.5); its derivative exists at the endpoints because $q > 1$. Equation (5.3) is zero at $p = p_b(q)$ and changes from negative to positive as p increases. At this p , strict increase of G' implies $G'(0) < G'(p^2) = 0$. Consequently $D_q(p_b(q)) > 0$, and Theorem 4.2 gives $p_b(q) < p_c(q)$. Positivity and the remaining bounds follow directly from the formulas. $\square$

Theorem 5.2 (Fixed-mean phase classification). *Fix a finite $q > 1$, and let $0 < s \leq 1$, $p = s/2$. Up to exchanging the two coins, the maximizer of either entropy at mean s is unique, and is as follows.*

- (i) If $0 < p \leq p_b(q)$, the maximizing pair is $(2p, 0)$.
- (ii) If $p_b(q) < p < p_c(q)$, the maximizing pair is $(p + \sqrt{u_*}, p - \sqrt{u_*})$, where $u_* \in (0, p^2)$ is the unique solution of

$$(A - u_*)^{q-1} + (C - u_*)^{q-1} = 2(B + 2u_*)^{q-1}. \quad (5.5)$$

- (iii) If $p_c(q) \leq p \leq 1/2$, the maximizing pair is (p, p) .

For $1 < s < 2$, complement the maximizing pairs at mean $2 - s$. At $s = 0$ and $s = 2$ the unique possible distributions are deterministic.

Proof. Minimize the strictly convex function $G_{q,p}$ on its compact interval. When $p \leq p_b(q)$, its right-end derivative is nonpositive, and its derivative is strictly negative before that endpoint. Its unique minimum is therefore at $u = p^2$. When $p \geq p_c(q)$, its left-end derivative is nonnegative, and its derivative is strictly positive thereafter, so its unique minimum is at $u = 0$. Between the boundaries, $G'(0) < 0 < G'(p^2)$; continuity and strict increase of G' give a unique interior zero, exactly (5.5). The only duplication in the map from u to parameter pairs comes from the two signs of t , which exchange the coins. Complement symmetry proves the remaining mean range. $\square$

The intermediate regime is important. Failure of concavity does not mean that the best distribution must always have a deterministic summand. There is an entire interval of means for which the unique unordered maximizer consists of two distinct, strictly interior Bernoulli parameters.

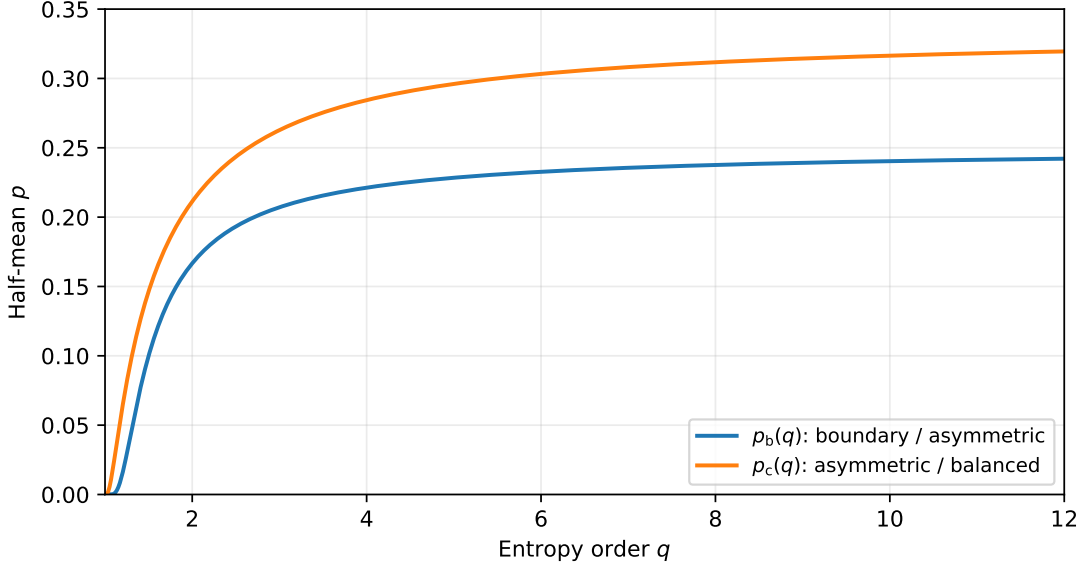


Figure 1: The exact boundaries (5.1) and (4.5), plotted numerically. Below the lower curve the optimizer is a boundary pair; between the curves it is an unequal interior pair; above the upper curve it is the balanced pair. The half-mean is $p = (p_1 + p_2)/2 \leq 1/2$. The plot illustrates Theorem 5.2; it is not used in its proof.

5.2 Closed formulas at order two

Let $s = p_1 + p_2$ and $r = p_1 p_2$. The probability vector is $(1 - s + r, s - 2r, r)$, so direct expansion gives

$$F_2(s, r) = 1 - 2s + 2s^2 + (2 - 6s)r + 6r^2 \quad (5.6)$$

$$= 6 \left(r - \frac{3s - 1}{6} \right)^2 + \frac{5 - 6s + 3s^2}{6}. \quad (5.7)$$

For $0 \leq s \leq 1$, the feasible product interval is $0 \leq r \leq s^2/4$. Thus the exact optimizer is

$$r_* = \max \left\{ 0, \min \left\{ \frac{s^2}{4}, \frac{3s - 1}{6} \right\} \right\}. \quad (5.8)$$

This gives the boundary regime $s \leq 1/3$, the unequal interior regime

$$\frac{1}{3} < s < 1 - \frac{1}{\sqrt{3}},$$

and the balanced regime above it. In the middle regime,

$$(p_1, p_2) = \frac{1}{2} \left(s \pm \sqrt{s^2 - 2s + \frac{2}{3}}, \quad s \mp \sqrt{s^2 - 2s + \frac{2}{3}} \right), \quad (5.9)$$

and the minimum collision probability equals $(5 - 6s + 3s^2)/6$. The notation in (5.9) means the two opposite sign choices, which exchange the coordinates.

On the path (3.1), the complete collision identity is

$$F_2(p + t, p - t) - F_2(p, p) = -2(1 - 6p + 6p^2)t^2 + 6t^4. \quad (5.10)$$

It supplies a second way to check the rational certificate: $p = 1/10$ and $t = 1/20$ give $F_2(p, p) - F_2(p + t, p - t) = 181/80000$.

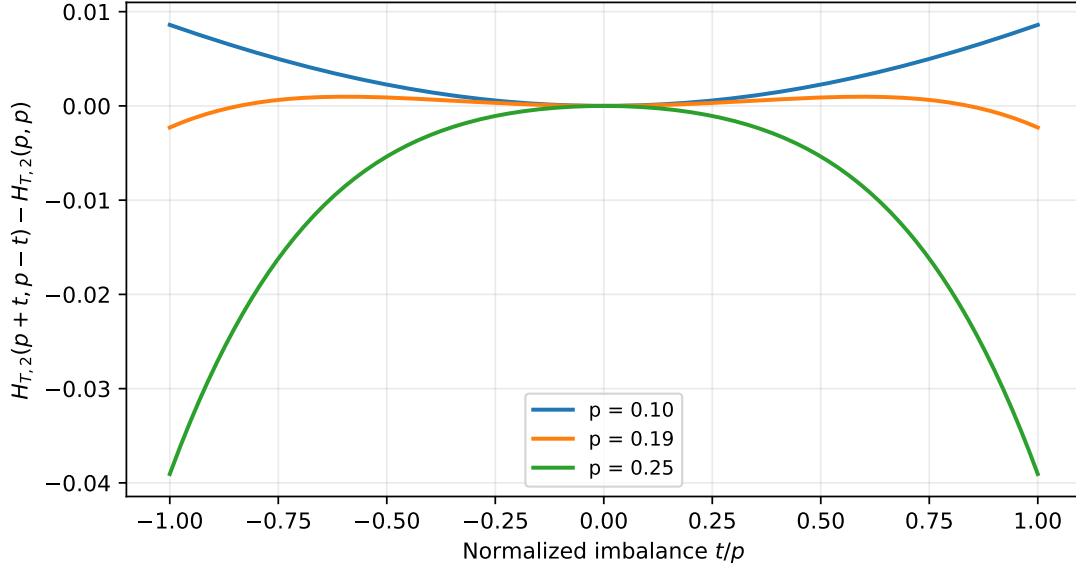


Figure 2: Exact order-two Tsallis entropy differences from the balanced value, evaluated from (5.10). The three half-means lie in the boundary, unequal-interior and balanced regimes, respectively. A positive value away from zero is a strict midpoint obstruction when the two symmetric parameter points are used as endpoints.

6 The Shannon transition and subcritical fixed-mean concavity

6.1 Monotonicity and asymptotics of the boundaries

Proposition 6.1. *Both $p_b(q)$ and $p_c(q)$ are strictly increasing for $q > 1$. With $a = q - 1$ and $L = \log 2$, as $a \downarrow 0$,*

$$p_b(1+a) = \frac{1}{2}e^{-L/a}(1 + O(e^{-L/a})), \quad (6.1)$$

$$p_c(1+a) = \exp\left(-\sqrt{\frac{2L}{a}} + O(\sqrt{a})\right). \quad (6.2)$$

As $a \rightarrow \infty$,

$$p_b(1+a) = \frac{1}{4} - \frac{L}{8a} + O(a^{-2}), \quad p_c(1+a) = \frac{1}{3} - \frac{2L}{9a} + O(a^{-2}). \quad (6.3)$$

Proof. Monotonicity of p_b follows directly from (5.1). Set $g(a) = \operatorname{arcosh}(e^{La})$, with $g(0) = 0$. For $a > 0$,

$$g'(a) = \frac{L}{\sqrt{1 - e^{-2La}}}, \quad g''(a) = -\frac{L^2 e^{-2La}}{(1 - e^{-2La})^{3/2}} < 0.$$

Strict concavity and $g(0) = 0$ imply that $g(a)/a$ strictly decreases. Equation (4.5) then makes p_c strictly increasing. For small a , the expansion $\cosh z = 1 + z^2/2 + O(z^4)$ gives $g(a) = \sqrt{2La}(1 + O(a))$. Substitution into (4.5) gives (6.2), while (6.1) is immediate. For large a ,

$$g(a) = (a+1)L + O(2^{-2a}).$$

Expanding the two explicit boundary formulas in $1/a$ gives (6.3). $\square$

The distinction between the two exponentially small scales is substantial. The explicit dyadic construction guarantees a finite witness without solving an equation, but is not close to the sharp

q	$p_b(q)$	$p_c(q)$
1.01	3.944305×10^{-31}	7.597324×10^{-6}
1.05	4.768367×10^{-7}	0.004986586
1.1	0.000487805	0.022610828
1.25	0.029411765	0.081406962
1.5	0.100000000	0.146446609
2	0.166666667	0.211324865
3	0.207106781	0.262751066
4	0.221246667	0.284370123
10	0.240377712	0.316443755
100	0.249124818	0.331779267

Table 1: Numerical boundary values from the exact formulas; the CSV in the archive retains more digits. For $q = 1.01$ the balanced-point instability requires a half-mean below about 7.6×10^{-6} , whereas the simple boundary-optimizer condition is far more restrictive.

local scale when q is near one. A coarse scan of moderate probabilities can therefore miss the instability near Shannon entropy. This is an explanation of the sampling issue, not a claim about how the original conjecture was tested.

6.2 A nonuniform limit of curvature

For any fixed $p \in (0, 1/2]$, expansion of (4.1) around $a = 0$ gives

$$D_{1+a}(p) = -2a \log 2 + O_p(a^2). \quad (6.4)$$

Indeed the coefficient of a is $2 \log(1-p) - 2 \log(2p(1-p)) + 2 \log p = -2 \log 2$. Consequently each of (4.3) and (4.4) approaches $-4 \log 2$ as $q \downarrow 1$.

In contrast, if $q > 1$ is fixed and $p \downarrow 0$, then $D_q(p) \rightarrow 1$ and $F_q(p, p) \rightarrow 1$, so both curvatures approach

$$\frac{2q}{q-1} > 0. \quad (6.5)$$

Thus convergence to the Shannon curvature is not uniform near the parameter boundary. There is no contradiction between the Shannon theorem and counterexamples at every order immediately above one.

6.3 A positive result below and at one

Although general joint concavity for $0 < q < 1$ is not established here, the fixed-mean two-coin problem admits a complete result.

Theorem 6.2 (Concavity on every two-coin fixed-mean segment). *For each $0 < q \leq 1$ and each mean $s \in (0, 2)$, Rényi and Tsallis entropy are strictly concave along the nondegenerate interior of the two-coin parameter segment $p_1 + p_2 = s$, using their common Shannon value at $q = 1$. They extend as concave functions to the closed segment. Their unique maximizer is the balanced pair $(s/2, s/2)$.*

Proof. By complement symmetry assume $s \leq 1$ and use $p = s/2$, $u = t^2$. First suppose $0 < q < 1$, so $a = q - 1 \in (-1, 0)$. The factorization (4.6) is still valid at $u = 0$. Since $2^a < 1$,

$$y^2 - 2^{a+1}y + 1 = (y - 2^a)^2 + 1 - 2^{2a} > 0.$$

Thus $D_q(p) > 0$ and $G'(0) = -qD_q(p) < 0$. Equation (5.4) now has a negative prefactor $q(q-1)$, so $G''(u) < 0$ throughout the interior. Hence $G'(u) < 0$ there. With $K(t) = G(t^2)$, the chain rule gives

$$K''(t) = 2G'(t^2) + 4t^2G''(t^2) < 0. \quad (6.6)$$

Tsallis entropy is $(K-1)/(1-q)$, so it is strictly concave. Rényi entropy is $(\log K)/(1-q)$, and $(\log K)'' = K''/K - (K'/K)^2 < 0$, giving the same conclusion. Evenness and strict decrease of G in u make $t = 0$ the unique maximum. Continuity of x^q at zero extends concavity to the closed segment.

For $q = 1$, write $J(u) = -\sum_{k=0}^2 f_k(u) \log f_k(u)$, with f_k as in (3.4). Direct differentiation gives

$$J'(u) = \log \frac{f_0(u)f_2(u)}{f_1(u)^2}, \quad (6.7)$$

$$J''(u) = -\frac{1}{f_0(u)} - \frac{4}{f_1(u)} - \frac{1}{f_2(u)} < 0. \quad (6.8)$$

For any two Bernoulli parameters v, w , the identity $f_1 = v(1-w) + w(1-v)$ and the arithmetic–geometric mean inequality imply $f_1^2 \geq 4f_0f_2$. Thus $J'(u) \leq -\log 4 < 0$. Applying the chain rule as in (6.6) proves strict concavity in t , and continuity of $x \log x$ at zero handles the endpoints. At $t = 0$, $f_1^2 = 4f_0f_2$, so the second derivative is exactly $-4 \log 2$. $\square$

This theorem concerns a one-dimensional family of directions in a *two*-dimensional parameter cube. It is not a proof of joint concavity on that cube for $q < 1$, much less for arbitrary numbers of coins.

7 Minimum dimension and separate coordinate concavity

7.1 Tsallis entropy remains concave in each coordinate

Proposition 7.1. *For every n and every finite $q > 0$, Tsallis entropy of a Bernoulli sum is concave in each parameter separately, with all other parameters fixed. Nevertheless it fails joint concavity when $n = 2$ and $q > 1$.*

Proof. Remove coin i , and let g_k be the probability mass function of the remaining sum, extended by zero outside its support. With $z = p_i$,

$$f_k(z) = (1-z)g_k + zg_{k-1}, \quad d_k = g_{k-1} - g_k.$$

For $q \neq 1$ and $0 < z < 1$, direct differentiation gives

$$\frac{d^2}{dz^2} H_{T,q} = -q \sum_k d_k^2 f_k(z)^{q-2} \leq 0. \quad (7.1)$$

Indices at which both g_k and g_{k-1} vanish are omitted; their contribution is identically zero. The formula for $q = 1$ is $-\sum_k d_k^2 / f_k(z) \leq 0$. Continuity extends the conclusion to $z = 0, 1$. The failure of joint concavity is Theorem 1.1. $\square$

In particular, one coin cannot be a Tsallis counterexample at any finite positive order. The minimum number of coins needed for failure is exactly two for every $q > 1$. Formula (4.9) gives a tangible matrix explanation: a symmetric matrix may have negative diagonal entries while possessing a positive eigenvalue.

7.2 The one-coin Rényi calculation

Proposition 7.2 (Minimum number of coins for Rényi failure). *For $1 < q \leq 2$, Rényi entropy of one Bernoulli variable is concave, so the minimum number of coins needed for a counterexample is two. For every finite $q > 2$, one coin already gives failure.*

Proof. Put $a = q - 1$, $b = 1 - z$, and $C(z) = z^q + b^q$. Then

$$H''_{R,q}(z) = -\frac{C''(z)C(z) - C'(z)^2}{aC(z)^2}.$$

For $0 < a \leq 1$, set $x = z/b > 0$ and $w = \log x$. Expanding the numerator gives

$$C''C - (C')^2 = qb^{2a}N_a(x), \quad (7.2)$$

$$N_a(x) = ax^{a-1}(1+x^2) + 2(a+1)x^a - x^{2a} - 1 \quad (7.3)$$

$$= 2x^a[a \cosh w + (a+1) - \cosh(aw)]. \quad (7.4)$$

Convexity of $\cosh$ gives $\cosh(aw) \leq a \cosh w + (1-a) \cosh 0$. Hence $N_a(x) \geq 4ax^a > 0$, proving strict concavity in the interior. Continuity gives concavity on $[0, 1]$.

If $q > 2$, as $z \downarrow 0$ we have $C \rightarrow 1$, $C' \rightarrow -q$, and $C'' \rightarrow q(q-1)$. Therefore

$$H''_{R,q}(z) \rightarrow \frac{q}{q-1} > 0.$$

This supplies a one-coin convex interval. Combining these facts with Theorem 1.1 proves the minimum-dimension statements. $\square$

For example, at $q = 2$ the one-coin identity simplifies to

$$H''_{R,2}(z) = -\frac{8z(1-z)}{(z^2 + (1-z)^2)^2} \leq 0.$$

The elementary one-coin obstruction above order two is consistent with the source's original test family; the two-coin antisymmetric direction is what forces the bound down to one.

8 Higher dimensions and structural consequences

8.1 Strictly interior counterexamples with any number of coins

Appending deterministic zero coins immediately transfers a two-coin counterexample to every larger dimension. A stronger statement avoids boundary parameters altogether.

Theorem 8.1 (Instability at small binomial parameter vectors). *Fix $n \geq 2$ and a finite $q > 1$. For all sufficiently small $p > 0$, each entropy has positive second variation at $(p, \dots, p)$ in the direction $(1, -1, 0, \dots, 0)$. Consequently every dimension $n \geq 2$ admits a strict interior midpoint counterexample of this form.*

Proof. Let $W \sim \text{Bin}(n-2, p)$ be the background sum, independent of the first two coins, and write $g_k = \mathbb{P}(W = k)$ with zero extension. Convolution (3.2) with g shows that

$$f_k(t) = f_k(0) - t^2(g_k - 2g_{k-1} + g_{k-2}), \quad (8.1)$$

where $f(0)$ is the $\text{Bin}(n, p)$ mass function. All first derivatives at zero vanish. Thus the Tsallis second variation is

$$\frac{2q}{q-1} \sum_{k=0}^n f_k(0)^{q-1} (g_k - 2g_{k-1} + g_{k-2}), \quad (8.2)$$

and the Rényi second variation is this quantity divided by $F_q(0) > 0$. As $p \downarrow 0$, the $k = 0$ summand in (8.2) tends to one. For every $k \geq 1$, $f_k(0)^{q-1} \rightarrow 0$, while the bracket is bounded. The finite sum therefore tends to one. Both second variations are positive for sufficiently small p . Choose $|t|$ sufficiently small compared with p . Smoothness gives a strict local entropy minimum at $t = 0$, and exchanging the first two coins makes the two nonzero endpoints have equal entropy. All coordinates are interior. $\square$

The proof is dimensionwise: the allowed size of p may depend on n and q . No uniform-in- n neighborhood is asserted.

8.2 Parameter majorization and the geometry of the counterexample

At a fixed small mean, averaging the two unequal parameter vectors in Theorem 3.2 *decreases* entropy. Thus neither entropy is Schur-concave as a function of the Bernoulli *parameters* for $q > 1$. Nor is it Schur-convex on the whole parameter square: at mean one, Theorem 4.2 shows that the balanced point $(1/2, 1/2)$ is a strict local maximum, reversing the required Schur-convex ordering. These are statements about parameter vectors, not about majorization of probability mass vectors.

There is an elementary distributional interpretation of the direction. Increasing $u = t^2$ transfers mass u from outcome zero and mass u from outcome two to outcome one. The mean stays fixed, while

$$\text{Var}(S) = 2p(1 - p) - 2u \tag{8.3}$$

decreases. When p is small, the mass at zero is overwhelmingly the largest. At orders above one, reducing that dominant mass can reduce the power sum more than the loss of a rare outcome increases it. The resulting entropy increase is therefore compatible with a decrease in variance. This interpretation is encoded exactly, rather than assumed heuristically, in (3.5).

8.3 What the result does not contradict

The Shannon joint-concavity theorem is untouched. So are positive results about monotone changes of one parameter, including the separate-coordinate Tsallis inequality (7.1). The later Hillion–Johnson question about increasing parameters towards $1/2$ is a monotonicity question [3, Section 4]; the present path has slopes of opposite signs and is not a counterexample to that question. Likewise, convolution entropy-power inequalities such as those studied in [4] are not Jensen inequalities in the Bernoulli parameters.

9 Verification, computation and reproducibility

9.1 Three independent layers

The mathematical argument is deliberately independent of a search program. The main disproof consists of the three-probability identity (3.2), the derivative bound in Lemma 3.1, and the explicit arithmetic estimate in Theorem 3.2. The rational order-two certificate can be checked without differentiation or logarithm evaluation.

The archive adds the following checks.

- (1) `code/verify_exact.py` uses only Python’s standard library and exact rational arithmetic. The recorded run passed **12,512 checks**: 24 probability calculations were compared with independent exhaustive outcome enumeration; the collision polynomial was tested on 441 rational parameter pairs; 1,991 integer-order Jensen examples, 16 rational-order sufficient-condition certificates, 11 higher-dimensional witnesses, and 9,999 fixed-mean optimizer

comparisons were checked. The remaining checks cover normalization and the main explicit certificate.

- (2) `code/verify_symbolic.py` independently differentiates symbolic expressions using SymPy. It checks the general center-curvature and quartic identities, the collision polynomial, the one-coin order-two formula and the exact Hessian (4.9).
- (3) `code/make_figures.py` evaluates the explicit boundaries at 80-decimal working precision and produces the numerical CSV and vector figures. These numerical illustrations play no role in the proof.

A floating-point Hessian search was used during exploration to find promising directions. Its script and log are retained for provenance, but no theorem relies on an eigenvalue returned by floating-point linear algebra. In particular, the exploratory log is not an exact certificate.

9.2 Rebuilding

From the archive’s top-level directory, run

```
python code/verify_exact.py
python code/verify_symbolic.py
python code/make_figures.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

Only the last two optional verification/illustration programs require third-party Python libraries; the exact certificate checker has none. A prebuilt PDF, the numerical table and the two figure PDFs are included, so rebuilding the article itself does not require running Python. The dependency versions used are recorded in `results/environment.txt`.

10 Conclusions and remaining questions

The proposed numerical generalized Shepp–Olkin thresholds are false in the unrestricted affine-interpolation setting. The failure is already visible with two independent coins, at a fixed mean, away from every face of the parameter square. It persists for every finite order above one and defeats even quasiconcavity. A single strict rational inequality at order two is sufficient to disprove both proposed values.

The counterexample is not merely isolated. Its sharp curvature boundary is (4.5); its fixed-mean maximizers undergo the three rigorously classified regimes of Theorem 5.2; and its approach to the Shannon order has the singular scale (6.2). Below one, Theorem 6.2 gives a positive result on every two-coin fixed-mean segment, but not on every affine direction.

The most immediate remaining mathematical question is whether universal joint concavity holds for all $0 < q < 1$. An affirmative result would complete a corrected finite-order threshold theorem with threshold one. A second, distinct question is what happens after restricting all interpolation slopes to the same sign. The present construction deliberately uses a mixed-sign direction and does not settle that restriction. A third is to classify fixed-mean entropy maximizers for three or more Bernoulli parameters; Theorem 8.1 rules out a universal balanced-maximizer principle above one, but is not such a classification.

Scope of the order parameter. All threshold and minimum-dimension assertions in this article concern finite positive orders. In particular, the direct limit of the Tsallis functional as $q \rightarrow \infty$ is the constant zero functional, since $0 \leq H_{T,q} \leq 1/(q-1)$. That limiting functional should not be

silently included in a finite-order threshold statement. No classification at order zero is asserted here.

Research status. The article presents self-contained proofs, not just computational suggestions, but it is an AI-assisted research draft. No independent referee report, author confirmation, or proof-assistant verification is claimed. The novelty claim is limited to what the documented literature search established. These limitations concern review and priority; they do not replace or weaken the explicit mathematical statements and proofs.

A Random area selection and manifest exclusion

The area list was fixed before a single call to Python’s `secrets.randbelow(96)`. The returned zero-based index was 57, hence the selected one-based area was 58: *Discrete probability*. The original draw is in `notes/selection.json`; its full generating script is in `code/select_area.py`. There was no redraw. Re-executing that script makes a *new* random draw; it does not reproduce the original OS randomness. The preserved JSON is the historical selection record. The area labels overlap, so uniformity is over this declared list of labels, not over an intrinsic partition of all mathematics.

The supplied `manifest(1).tex`, dated 20 September 2026, catalogues 71 research packages. All entries were read as exclusion targets, without attempting to revalidate their claimed proofs. None concerns the numerical Rényi/Tsallis thresholds, the two-coin fixed-mean problem, or the particular Hillion–Johnson conjecture selected here. The exclusion audit is recorded in `notes/manifest_exclusion.md`.

The complete 96-area list

- | | | |
|--------------------------------|--------------------------------|-------------------------------|
| 1. Additive combinatorics | 25. Tiling theory | 49. Knot theory |
| 2. Analytic number theory | 26. Combinatorics on words | 50. Low-dimensional topology |
| 3. Algebraic number theory | 27. Automata theory | 51. Differential geometry |
| 4. Diophantine approximation | 28. Formal language theory | 52. Metric geometry |
| 5. Finite fields | 29. Semigroup theory | 53. Geometric group theory |
| 6. Arithmetic dynamics | 30. Finite group theory | 54. Dynamical systems |
| 7. Combinatorial number theory | 31. Infinite group theory | 55. Symbolic dynamics |
| 8. Integer partitions | 32. Combinatorial group theory | 56. Ergodic theory |
| 9. Permutation patterns | 33. Ring theory | 57. Probability inequalities |
| 10. Algebraic combinatorics | 34. Commutative algebra | 58. Discrete probability |
| 11. Enumerative combinatorics | 35. Noncommutative algebra | 59. Random walks |
| 12. Extremal set theory | 36. Representation theory | 60. Stochastic processes |
| 13. Ramsey theory | 37. Lie algebras | 61. Special functions |
| 14. Spectral graph theory | 38. Universal algebra | 62. Orthogonal polynomials |
| 15. Structural graph theory | 39. Lattice theory | 63. Continued fractions |
| 16. Graph polynomials | 40. Order theory | 64. Asymptotic analysis |
| 17. Graph labeling | 41. Category theory | 65. Complex analysis |
| 18. Design theory | 42. Type theory | 66. Geometric function theory |
| 19. Coding theory | 43. Proof theory | 67. Harmonic analysis |
| 20. Matroid theory | 44. Computability theory | 68. Functional analysis |
| 21. Combinatorial geometry | 45. Model theory | 69. Banach space geometry |
| 22. Discrete geometry | 46. Set-theoretic topology | 70. Operator theory |
| 23. Convex geometry | 47. General topology | 71. Matrix inequalities |
| 24. Lattice polytopes | 48. Algebraic topology | 72. Tensor algebra |

- | | | |
|---------------------------------|---------------------------------------|---------------------------|
| 73. Optimization theory | 81. Hypergraph theory | 89. Symmetric functions |
| 74. Discrete optimization | 82. Incidence geometry | 90. q-series |
| 75. Numerical analysis | 83. Finite geometry | 91. Difference equations |
| 76. Approximation theory | 84. Topological combinatorics | 92. Functional equations |
| 77. Combinatorial game theory | 85. Cluster algebras | 93. Fractional calculus |
| 78. Mathematical logic of games | 86. Tropical geometry | 94. Integral inequalities |
| 79. Information theory | 87. Algebraic statistics | 95. Potential theory |
| 80. Boolean functions | 88. Combinatorial commutative algebra | 96. Fractal geometry |

B A compact independent certificate

The following Python fragment is a complete exact verification of the order-two Jensen obstruction. It uses neither the derivative formulas nor any third-party package.

```
from fractions import Fraction as Q

def collision(a, b):
    f = ((1-a)*(1-b), a+b-2*a*b, a*b)
    return sum(x*x for x in f)

center = collision(Q(1,10), Q(1,10))
endpoint = collision(Q(3,20), Q(1,20))
assert center == Q(55088,80000)
assert endpoint == Q(54907,80000)
assert center - endpoint == Q(181,80000) > 0
```

Both endpoint entropies are equal by exchanging the coins. For Tsallis order two, subtracting the power sum from one reverses the displayed inequality. For Rényi order two, applying the decreasing function $-\log$ reverses it. Thus the fragment certifies a genuine Jensen failure, not just a numerical approximation to one.

References

- [1] E. Hillion and O. Johnson, *A proof of the Shepp–Olkin entropy concavity conjecture*, Bernoulli **23**(4B) (2017), 3638–3649. [doi:10.3150/16-BEJ860](https://doi.org/10.3150/16-BEJ860). Original version: [arXiv:1503.01570v1](https://arxiv.org/abs/1503.01570v1) (5 March 2015), especially Section 4 and Conjecture 4.2 on PDF page 7.
- [2] O. Johnson, *Entropy and thinning of discrete random variables*, The IMA Volumes in Mathematics and its Applications **161** (2017), 33–53. [arXiv:1510.05390v3](https://arxiv.org/abs/1510.05390v3) (7 June 2016), Section 8, item 5.
- [3] E. Hillion and O. Johnson, *A proof of the Shepp–Olkin entropy monotonicity conjecture*, Electronic Journal of Probability **24** (2019), paper 126, 1–14. [arXiv:1810.09791](https://arxiv.org/abs/1810.09791). The Rényi and Tsallis monotonicity discussion is in Section 4.
- [4] M. Madiman, J. Melbourne and C. Roberto, *Bernoulli sums and Rényi entropy inequalities*, Bernoulli **29**(2) (2023), 1578–1599. [doi:10.3150/22-BEJ1511](https://doi.org/10.3150/22-BEJ1511). [arXiv:2103.00896](https://arxiv.org/abs/2103.00896); the version consulted was v2, dated 15 March 2024.